

Question 3

Describing the word "ideal" in a semiconducting device

⇒ The term "ideal" implies that the device behaves perfectly under theoretical conditions. (eg. no leakage current, perfect mobility of charge carriers). In real-world applications, factors like temperature, impurities, and physical defects as affect performance.

Question 4

a) Determine the thermal voltage for a diode current at a temperature of 25°C .

b) For the same diode of part (a), find the diode current using I_D equation if $I_S = 40 \text{ nA}$, $n = 2$. (low value of V_D) and the applied bias voltage is 0.5 V .

$$I_D = I_S (e^{V_D / nV_T} - 1) \quad V_T = \frac{kT}{q}, \quad T_K = T_C + 273^{\circ}$$

$$K = 1.38 \times 10^{-23} \text{ J/K}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$T = 298 \text{ K}$$

$$V_T = \frac{(1.38 \times 10^{-23}) \times (298)}{1.6 \times 10^{-19}} = \underline{\underline{0.0259 \text{ V}}}$$

$$V_D = nV_T \ln\left(\frac{I_D}{I_S} + 1\right)$$

$$I_D = 40 \times 10^{-9}$$

$$I_S = 40 \times 10^{-9}$$

$$\underline{\underline{V_D = 0.5 \text{ V}}}$$

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